

The Convolution Operation

Deep Learning

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Contents of This Video

In this video, we will cover:

- The convolution operation step by step
- How filters detect features
- Stride: how far the filter moves
- Padding: handling image borders
- Output size calculation

The Convolution Operation

Convolution: Element-wise Multiply, Then Sum

1	2	0	1	2
0	1	3	2	1
2	3	1	0	2
1	0	2	1	0
0	1	1	2	1

Input (5×5)

×

1	0	-1
1	0	-1
1	0	-1

Filter (3×3)

=

$$\begin{aligned}
 &1 \times 1 + 2 \times 0 + 0 \times (-1) \\
 &+ 0 \times 1 + 1 \times 0 + 3 \times (-1) \\
 &+ 2 \times 1 + 3 \times 0 + 1 \times (-1)
 \end{aligned}$$

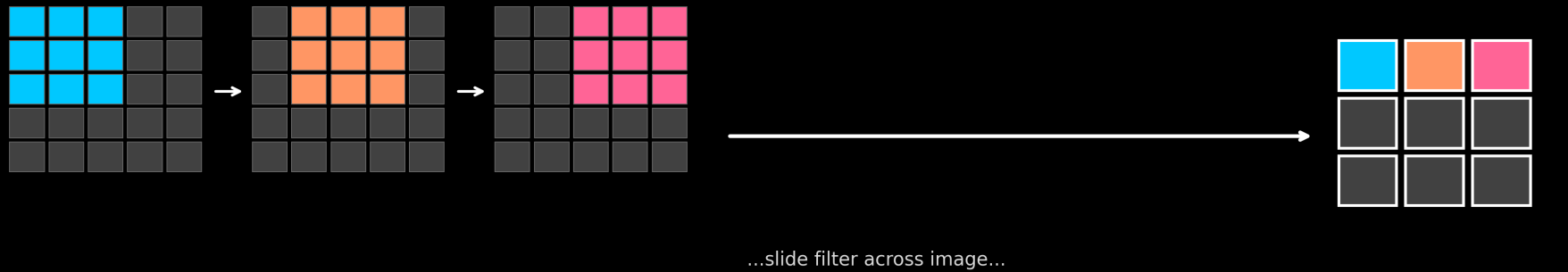
$$= -1$$

-1

One output

Sliding the Filter

Filter Slides Across Image, Producing Feature Map



Filters Detect Features

Vertical Edge
Detector

-1	0	1
-1	0	1
-1	0	1

Horizontal Edge
Detector

-1	-1	-1
0	0	0
1	1	1

Averaging
(Blur)

$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$
$1/9$	$1/9$	$1/9$

Different filters detect different features!

Stride: How Far to Move

Stride = how many pixels the filter moves each step

Stride 1:

- Move one pixel at a time
- Output slightly smaller than input
- More overlap, more computation

Stride 2:

- Move two pixels at a time
- Output roughly half the input size
- Less overlap, less computation

Higher stride = smaller output (downsampling)

Padding: Handling Borders

Problem: Output is smaller than input (borders are lost)

Solution: Add zeros around the image before convolving

Valid padding (no padding):

- Output smaller than input
- 5×5 input, 3×3 filter $\rightarrow 3 \times 3$ output

Same padding:

- Pad so output = input size
- 5×5 input, 3×3 filter $\rightarrow 5 \times 5$ output
- Adds 1 pixel of zeros around border

Output Size Formula

$$\text{Output} = \left\lfloor \frac{\text{Input} + 2 \times \text{Padding} - \text{Filter}}{\text{Stride}} \right\rfloor + 1$$

Example: 32×32 input, 3×3 filter, stride 1, no padding

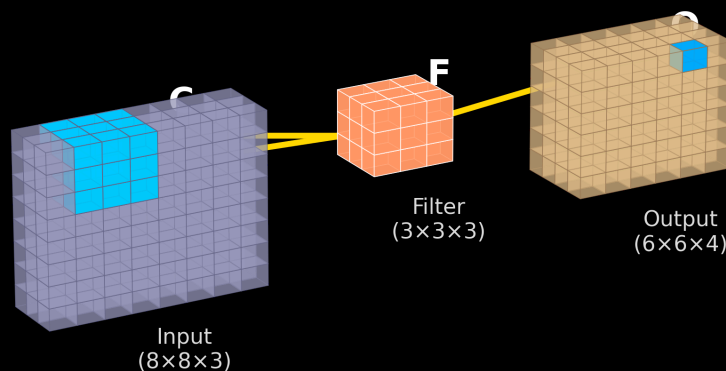
$$\text{Output} = \frac{32 + 0 - 3}{1} + 1 = 30$$

Example: 32×32 input, 3×3 filter, stride 1, same padding (pad=1)

$$\text{Output} = \frac{32 + 2 - 3}{1} + 1 = 32$$

Multiple Filters = Multiple Feature Maps

Convolution: $\text{Input} \times \text{Filter} \rightarrow \text{Output Volume}$



Each filter produces one feature map $\rightarrow$ K filters create depth K in output

What We've Covered

In this video, we've learned:

- Convolution: slide filter over image, multiply and sum at each position
- Filters detect features (edges, textures, etc.)
- Stride controls how far the filter moves (higher = smaller output)
- Padding adds zeros to control output size
- Output size = $(\text{Input} + 2 \times \text{Padding} - \text{Filter}) / \text{Stride} + 1$
- Multiple filters create multiple feature maps